



# High Speed High Resolution Current Comparator and its Application to Analog to Digital Converter

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**Abstract** This paper introduces a high speed high resolution current comparator which includes the current differencing stage and employs non linear feedback in the gain stage. The usefulness of the proposed comparator is demonstrated by implementing a 3-bit current mode flash analog-to-digital converter (ADC). Simulation program with integrated circuit emphasis (SPICE) simulations have been carried out to verify theoretical proposition and performance parameters of both comparator and ADC are obtained using TSMC 0.18  $\mu\text{m}$  CMOS technology parameters. The current comparator shows a resolution of  $\pm 5$  nA and a delay of 0.86 ns for current difference of  $\pm 1$   $\mu\text{A}$ . The impact of process variation on proposed comparator propagation delay has been studied through Monte Carlo simulation and it is found that percentage change in propagation delay in best case is 1.3 % only and in worst case is 9 % only. The ADC exhibits an offset, gain error, differential nonlinearity (DNL) and integral nonlinearity (INL) of 0.102  $\mu\text{A}$ , 0.99,  $-0.34$  LSB and 0.0267 LSB, respectively. The impact of process variation on ADC has also been studied at different process corners.

**Keywords** Current mode · Comparators · Nonlinear feedback · ADC

## Introduction

Current comparators are used in wide range of applications such as high speed data converters [1], winner-take-all circuits which are integral components of VLSI neural network implementations, fuzzy system and non linear filters [2], sequence detection [3], implantable biomedical devices [4] etc. The prime concern in all these applications is high resolution, high speed and low power consumption. Several current comparators are available in the literature [5–15]. The first current comparator [5] uses high output resistance current mirrors connected as a class AB stage to amplify small differences in input currents to large variations in output voltage. The high output resistance deteriorates the frequency performance when the inverter stages are added for rail to rail output swing. A source follower input stage is used in [6] which offers low input resistance and uses positive voltage feedback to amplify small voltage variations at the input stage node. It shows better performance than [5] but has a drawback of dead band region. A complicated biasing scheme is proposed in [7] to minimise the width of the dead band region, which on the downside increases the power dissipation. The concept of non linear feedback has been generalized in [8] and comparator in [6] has been shown as a specific example. The comparator in [9] adds two diode configured transistors in [6] to prevent the input source follower from entering in deep subthreshold state. As a result the input stage promptly follows the variations in input current, thereby improving the delay but limits the output voltage swing. The comparators [10, 11] use resistive feedback around an inverting amplifier stage to lower input and output impedances and improvement in response time. The speed and resolution in [10] is bias current dependant and thus, larger bias is needed for high resolution and speed

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which causes more power dissipation. A method [12] for achieving lower power consumption whilst maintaining high speed as in [6] is suggested where transistors are added to the input source follower stage. Current comparators [13, 14] are built using active blocks current conveyor (second generation) (CCII) [13] and differential current conveyor (second generation) (DCCII) [14] respectively. Some of the current comparators include current differencing stage [4, 5, 10, 13] whereas other assume input as the current difference [6–9, 11, 12, 15]. As the current differencing stage also contributes to the overall delay and power dissipation of current comparators, the performance of the circuits [6–9, 11, 12, 15] will deviate from the one reported therein.

A new current comparator is proposed in this paper. It includes a current difference stage, a gain stage with non linear feedback and an output stage. The current difference is generated using current mirror structure while a CMOS inverter is used as the output stage. To demonstrate the application of the proposed comparator, a 3-bit current mode flash ADC is implemented for an input current range 0 to 3.5  $\mu\text{A}$ . The performance of both comparator and ADC is studied through extensive SPICE simulations.

## Current Comparator

### Basic Concepts

The basic functionality of a current comparator is to determine which of the two input currents ( $I_{in1}$  or  $I_{in2}$ ) is greater and to present that decision as one of two voltage levels, established by the output's ( $V_{out}$ ) limiting values. The output value is typically interpreted as logic '1' and '0' for further processing. One of the two currents is generally a constant current called the reference current ( $I_{in2}$ ) against which the input current is compared ( $I_{in1}$ ). This is shown by the following relation:

$$V_{out}(t) = \begin{cases} 1, & I_{in1}(t) > I_{in2}(t) \\ 0, & I_{in1}(t) < I_{in2}(t) \end{cases} \quad (1)$$

The performance of the current comparator is determined in terms of resolution (minimum current difference it responds to) and delay which is defined as time required for output to switch between the two logic states following an input edge. The power dissipation and power delay product of the comparator are equally important which concerns low power applications. In the following sub section, the operating principle of the proposed comparator is discussed followed by the calculation of the above mentioned properties which are presented in the simulation section.

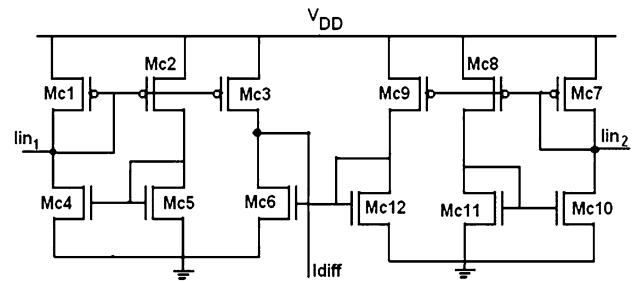


Fig. 1 Current difference stage

### Operating Principle

Figure 1 shows the current difference stage [14] of the proposed current comparator, with  $I_{diff}$  equal to the difference between input currents  $I_{in1}$  and  $I_{in2}$ . The identical current mirrors are used to replicate input currents into the transistor pair Mc3–Mc6, which then generates the current difference ( $I_{diff}$ ). This stage has an advantage of having low input impedance at input nodes  $I_{in1}$  and  $I_{in2}$ . Neglecting the output impedance of the transistors, the input impedances are respectively given by:

$$R_{in1} \approx \frac{1}{g_{mc1}} \quad (2)$$

$$R_{in2} \approx \frac{1}{g_{mc7}} \quad (3)$$

where  $g_{mc1}$  and  $g_{mc7}$  are transconductance of transistors Mc1 and Mc7 respectively.

The gain stage and the output stage of the proposed comparator are shown in Fig. 2. Here, the gain stage is transimpedance stage which consists of cascade of two resistive amplifiers (Mr1–Mr2, Mr3–Mr4) and a CMOS inverter (M1–M2) within the feedback loop (Mpf–Mnf). It converts the difference current  $I_{diff}$  sensed at node '1' to equivalent output voltage at node '2'. The transistors Mpf–Mnf form a non linear feedback around the gain stage, with their sources connected to node '1' while their gates are controlled by node '2'. This arrangement is similar to [6], where the input current difference node is source coupled with feedback around a CMOS inverter acting as a gain stage. However, the gain stage of the proposed comparator has higher gain within the feedback loop than [6] which culminates in better resolution and higher speed [8]. For very low input currents, the input and the output voltage of the gain stage are close to the quiescent point 'Q' and both Mpf and Mnf are OFF making node '1' capacitive which causes slow charging/discharging of the node voltage to which the gain and the output stages respond. For high positive values of current  $I_{diff}$ , node '1' voltage increases causes reduction in node '2' voltage due to inverting nature of cascaded amplifier stage. The transistor Mpf switches

ON as its source (node ‘1’) and gate (node ‘2’) voltages are respectively high and low, thus closing a negative feedback loop around the amplifier. A similar situation occurs for negative  $I_{\text{diff}}$  current where Mnf becomes ON. The transistors Mnf–Mpf are sized such that voltage excursions at node ‘1’ remain around quiescent point (Q-point) of the first resistive amplifier pair. This ensures a faster response since the gain is the maximum around the Q-point. The same principle is used to set the Q-point of Mr3–Mr4 and M1–M2. The CMOS inverter (M3–M4) forms the output stage and provides rail to rail swing.

The proposed current comparator is characterized on the basis of resolution, speed of operation, power dissipation and power delay product which are shown in the following section.

### Simulation Results

The functionality and circuit performance of the proposed comparator is verified through SPICE simulations using TSMC 0.18  $\mu\text{m}$  CMOS technology parameters with 1.8 V supply voltage. The dimensions of the transistors are enlisted in Table 1. The comparators [6–9, 11, 15] call the difference of the actual input current and the reference current as input current. The proposed comparator includes

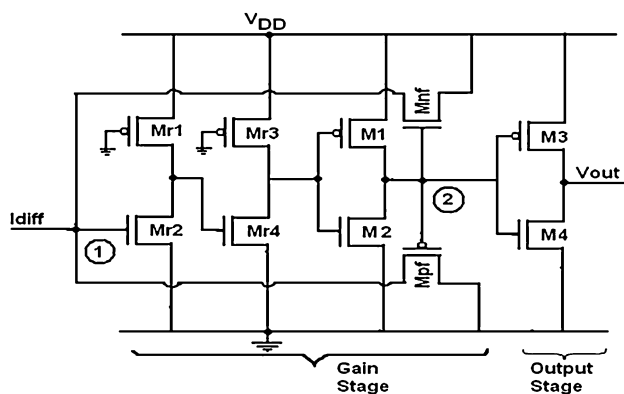


Fig. 2 Gain and output stages of the proposed comparator

the current difference circuit so both reference and input currents have to be applied. The terminal  $I_{\text{in1}}$  is used as the input current terminal and  $I_{\text{in2}}$  is referred to as the reference current ( $I_{\text{ref}}$ ) terminal.

The resolution, delay, power dissipation and power delay product (PDP) is determined through transient analysis. The input current  $I_{\text{in}}$  is varied by  $\pm\Delta I$  (1 nA to 1  $\mu\text{A}$ ) with respect to  $I_{\text{ref}} = 1 \mu\text{A}$ . A satisfactory operation of circuit is obtained till a minimum  $\Delta I$  of  $\pm 5$  nA, so the resolution of the proposed comparator is  $\pm 5$  nA. The response of comparator for  $\Delta I = \pm 100$  nA is shown in Fig. 3. The effect of variation in current difference on the delay and the power delay product (PDP) is also studied and the simulation

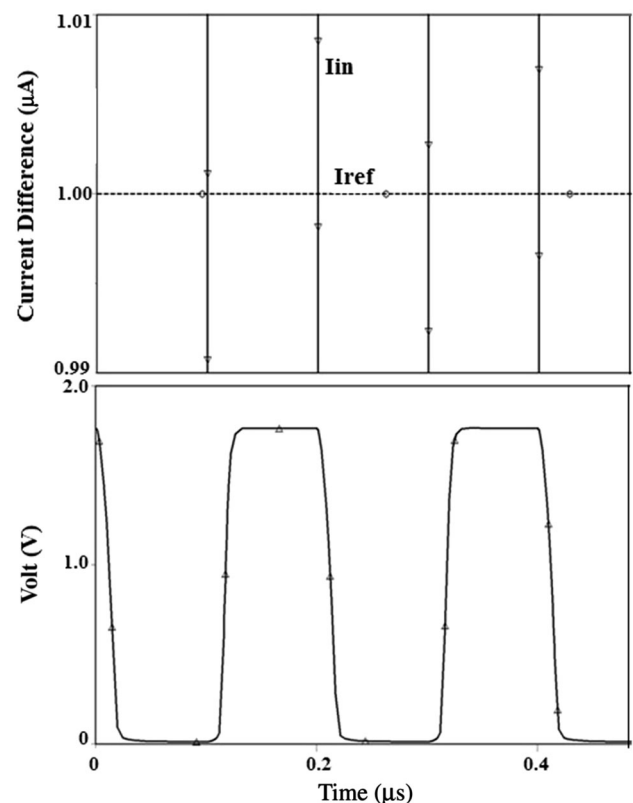
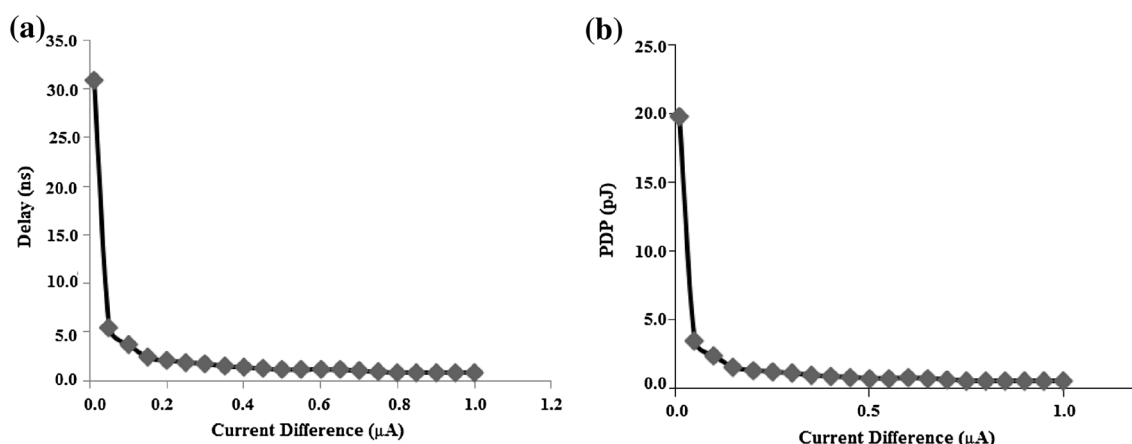


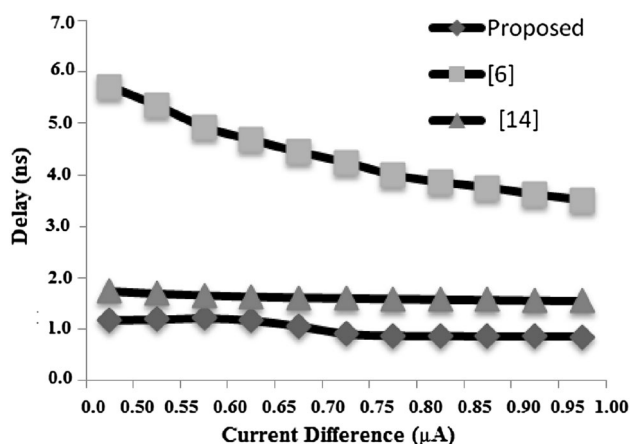
Fig. 3 Response of the comparator for current difference,  $\Delta I = \pm 100$  nA

Table 1 Comparator transistor dimensions

| Transistor         | Width ( $\mu\text{m}$ ) | Length ( $\mu\text{m}$ ) | Transistor | Width ( $\mu\text{m}$ ) | Length ( $\mu\text{m}$ ) |
|--------------------|-------------------------|--------------------------|------------|-------------------------|--------------------------|
| Mc1–Mc2, Mc4–Mc5   | 0.42                    | 0.18                     | Mc3–Mc6    | 0.66                    | 0.72                     |
| Mc7–Mc8, Mc10–Mc11 | 0.42                    | 0.18                     | Mc9–Mc12   | 0.66                    | 0.72                     |
| Mr1                | 0.4                     | 0.36                     | Mr2        | 1.677                   | 0.36                     |
| Mr3                | 0.4                     | 0.18                     | Mr4        | 19                      | 0.18                     |
| Mpf                | 0.54                    | 0.18                     | Mnf        | 0.54                    | 0.18                     |
| M1–M3              | 0.72                    | 0.36                     | M2–M4      | 0.42                    | 0.36                     |



**Fig. 4** Variation of **a** delay **b** pdp with current difference

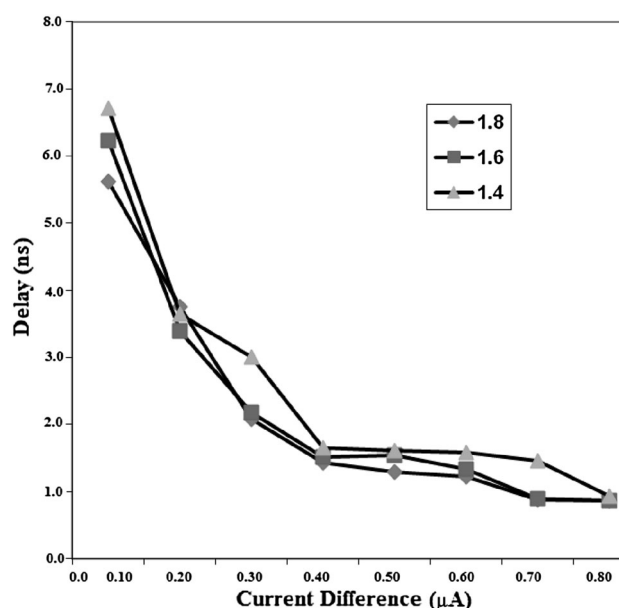


**Fig. 5** Delay comparison of the proposed comparator with those given in [6, 14]

results are depicted in Fig. 4. The power dissipation remains almost constant at 0.646 mW.

The suitability of the comparator is evaluated for current mode ADC wherein an input current is compared with respect to a constant reference current. The simulation conditions are kept same for fair comparison of the proposed comparator with those proposed in [6, 14]. The simulations show that proposed comparator has a resolution of  $\pm 5$  nA, whereas [6, 14] show  $\pm 50$  nA resolution. The delay is plotted in Fig. 5 for current difference in the range of  $\pm 0.5$  to  $\pm 1$   $\mu$ A. The proposed comparator clearly outperforms those given in [6, 14].

The circuit is also simulated at lower power supplies of 1.4 and 1.6 V. The simulated delay as a function of power supply voltage and the current difference is depicted in Fig. 6. It may be noted that the delay decreases with increase in power supply voltage which agrees with the established fact of inverse dependence between the two [17]. The impact of variations in oxide thickness ( $T_{ox}$ ), doping concentration ( $N_{ch}$ ), threshold voltage ( $V_{th0}$ ) and



**Fig. 6** Delay with respect to current difference and different power supplies

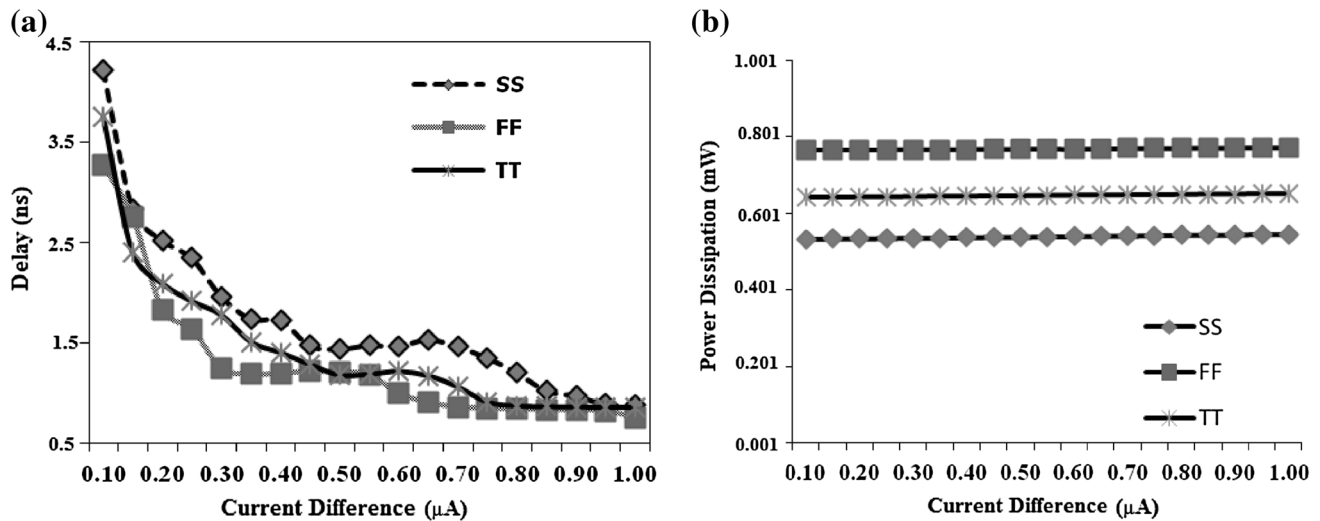
junction capacitance ( $C_J$ ) on propagation delay of the said comparator is shown in Table 2. To simulate this, Monte Carlo simulation was carried out for 50 runs for parameters  $T_{ox}$ ,  $N_{ch}$ ,  $V_{th0}$  and  $C_J$  with device variation of 0.05 % Gaussian distribution and lot variation of 0.05 % uniform distribution.

From the results, it is observed that the percentage change between median and nominal value is maximum (worst value) of 9 % only for  $I_{in} - I_{ref} = \pm 100$  nA and minimum of 1.3 % only for  $I_{in} - I_{ref} = \pm 1000$  nA.

The impact of parameter variation on delay and power dissipation of proposed comparator is studied at extreme process corners that is slow–slow (SS) and fast–fast (FF) corners and the results are shown in Fig. 7 along with the typical corner for comparison purpose. It may be noted

**Table 2** Variation in propagation delay in Monte Carlo analysis

| Propagation delay                | Nominal value (ns) | Mean (ns) | Median (ns) | Standard deviation (ns) |
|----------------------------------|--------------------|-----------|-------------|-------------------------|
| $I_{in} - I_{ref} = \pm 100$ nA  | 3.76               | 4.85      | 4.1         | 3.453                   |
| $I_{in} - I_{ref} = \pm 500$ nA  | 1.18               | 1.26      | 1.25        | 0.11                    |
| $I_{in} - I_{ref} = \pm 1000$ nA | 0.86               | 0.85      | 0.848       | 0.011                   |

**Fig. 7** Impact of parameter variation on **a** delay and **b** power dissipation against current difference

from Fig. 7a that the delay of the comparator for SS corner remains larger than typical (TT) corner case. Conversely, the delay under the FF process corner is overall lower than the delay in the TT due to faster charging/discharging of the parasitic capacitances. Owing to the same dependence, there is increased power dissipation for FF process corner than TT corner. Conversely, the power dissipation reduces for the SS case as compared to the TT case. This is in agreement with the results shown in Fig. 7b.

### Application Example

As an application of the proposed comparator, a 3-bit current mode flash ADC, as shown in Fig. 8 is implemented wherein  $I_{in}$  and  $I_{refi}$  ( $i = 1, \dots, 7$ ) represent input current and the reference currents of the  $i$ th comparator. The value of the  $I_{refi}$  is determined by the input current range and the number of bits ( $n$ ) in ADC output using the following relationship

$$I_{refi} = \frac{\text{Input Current Range}}{2^n - 1} i \quad (4)$$

While the current comparators receive progressively increasing currents as per Eq. (4), the input current ( $I_{in}$ ) is mirrored to all the current comparators that is all

comparators receive the same input current. Each comparator compares this input current to its respective reference current  $I_{refi}$ . Hence all comparators perform the comparison in a single step parallelly, so this structure is also called a parallel ADC or a single step ADC.

The output of the comparators represent thermometer code which gets converted into binary code ( $B_2 B_1 B_0$ ) by encoder block. The encoder is built using transmission gates to implement the logic given as

$$B_0 = C_1 \oplus C_2 \oplus C_3 \oplus C_4 \oplus C_5 \oplus C_6 \oplus C_7 \quad (5)$$

$$B_1 = \overline{C_4} C_2 + C_4 C_6 \quad (6)$$

$$B_2 = C_4 \quad (7)$$

where  $\overline{C_4}$  represents the complement of  $C_4$ .

The input current range for 3-bit ADC is chosen as 0 to 3.5 μA so the reference currents  $I_{refi}$ 's are set as 0.5 μA using Eq. (4). The functionality is verified through simulations using TSMC 0.18 μm CMOS technology parameters. The ideal and the actual ADC transfer characteristics are plotted using MATLAB and are shown in Fig. 9. The parameters such as offset, gain error, INL and DNL, are calculated from actual ADC transfer characteristic.

The offset is defined as the shift in the ADC transfer curve and the gain error as the difference between the slope of the lines through the end points of the actual and ideal

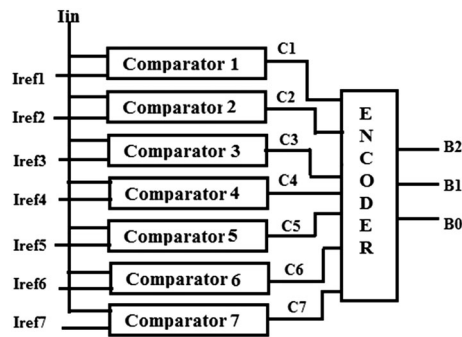


Fig. 8 3-bit current mode flash ADC

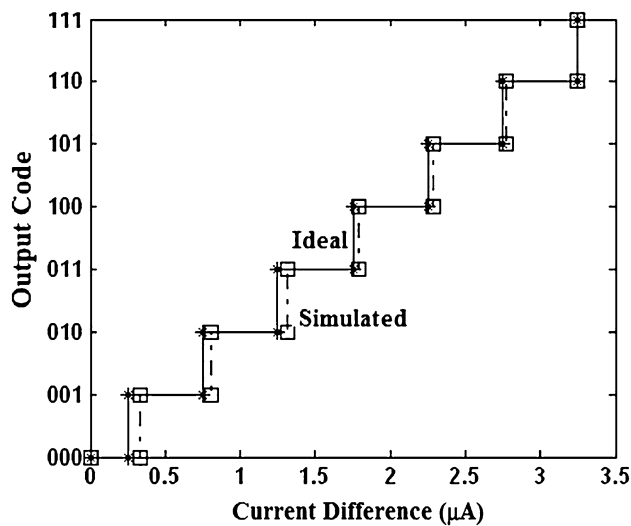


Fig. 9 Ideal and simulated ADC transfer characteristics

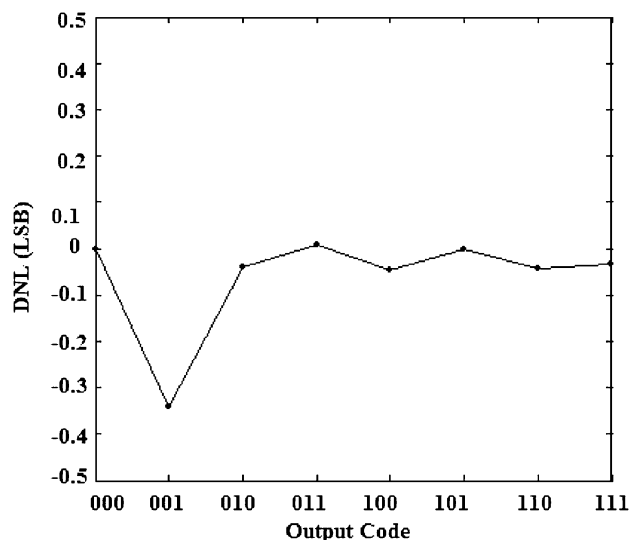


Fig. 10 DNL against output code

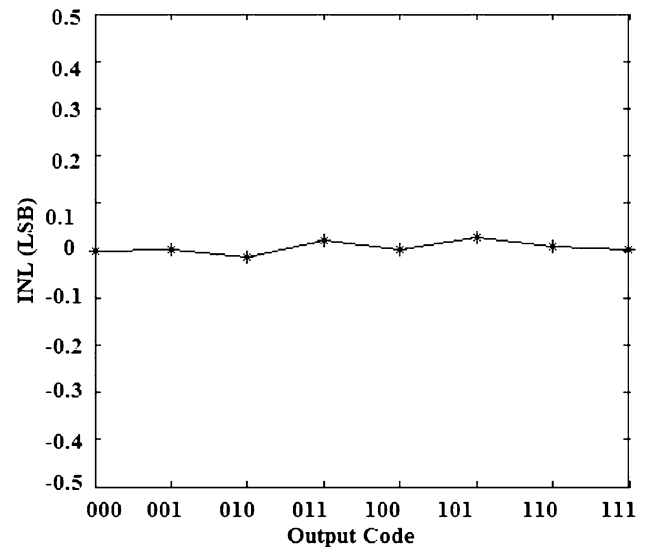


Fig. 11 INL against output code

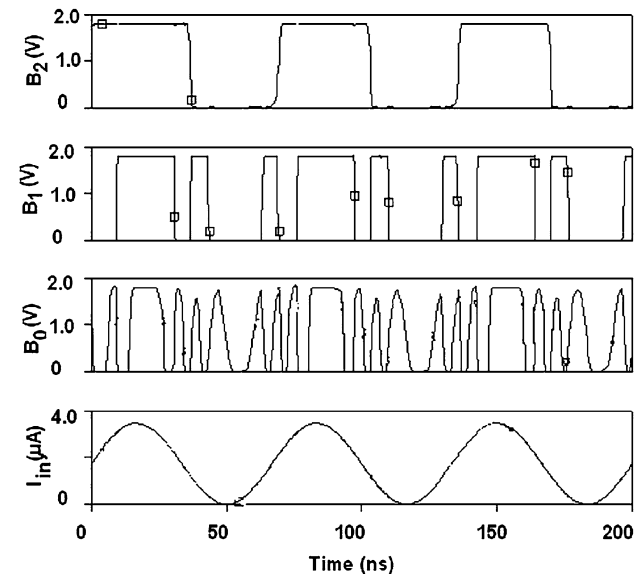
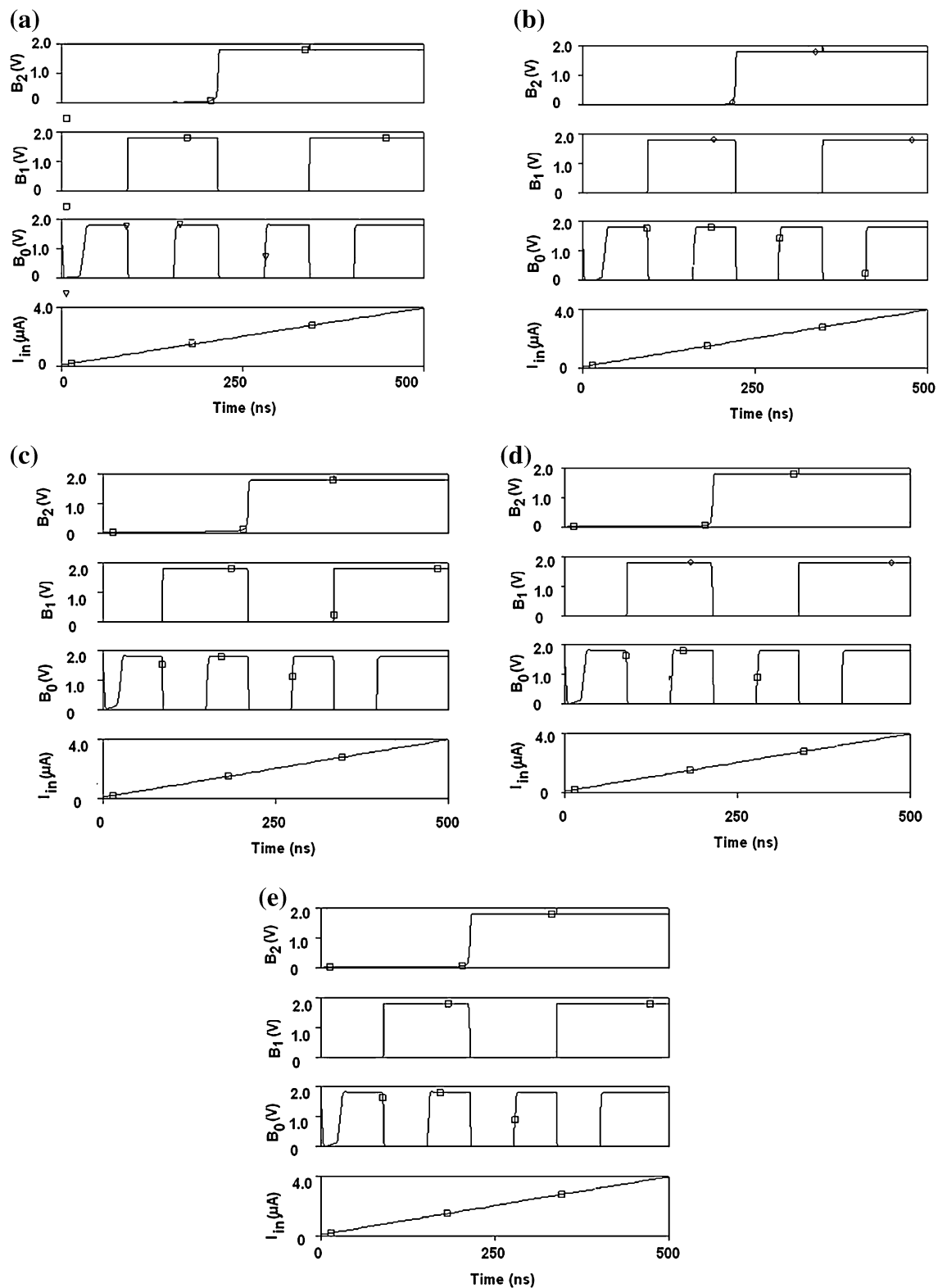


Fig. 12 ADC output against sine input at 15 MHz

ADC transfer curve. The offset and gain errors were found to be  $0.102 \mu\text{A}$  and  $0.99$ , respectively. The non linearity in the ADC output is measured in terms of integral non linearity (INL) and differential non linearity (DNL). The DNL at the  $n$ th step ( $\text{DNL}_n$ ) is defined as difference between actual width of the code transition at  $n$ th step and ideal step width. The INL is defined as the difference between input value for code  $n$  and the input value of the reference line at that code, where the reference line is the line drawn between the endpoints [16, 17].

Figure 9 is used to calculate DNL for each step using MATLAB and is plotted in Fig. 10. The INL was also



**Fig. 13** ADC Output for the process corners **a** TT, **b** SS, **c** FF, **d** SF **e** FS

calculated for each code using MATLAB is plotted in Fig. 11. The worst case DNL and INL are found to be  $-0.34$  LSB and  $0.0267$  LSB, respectively. To find the

maximum frequency of operation of the ADC, sine wave at different frequencies were applied to the ADC and the output of the ADC was observed in time domain. It was



found that the performance of the ADC is satisfactory up to a maximum frequency of 15 MHz and the corresponding input–output are shown in Fig. 12. The impact of process variation on the ADC behaviour is also studied. The ADC was simulated under different process corners namely SS, FF, SF and FS process corners under same simulation conditions as above and results are shown in Fig. 13. It can be seen that the ADC functions correctly across the process corners.

## Conclusion

A current comparator is proposed in this paper which has a resolution of  $\pm 10$  nA and has a propagation delay of 0.86 ns at input current of  $\pm 1$   $\mu$ A. The comparator uses non linear feedback to achieve high resolution and speed. The comparator's performance under reduced power supply voltage is also studied. The impact of process corners on delay and power dissipation of the proposed comparator was also studied. As an application of the proposed comparator, a 3-bit current mode flash ADC is implemented and performance is verified through SPICE simulations. It was found that the ADC functions correctly for various process corners.

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